

VATREEL · REQUIREMENTS

VatReel requirements

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DOCUMENT CONTROL

Document ID	VTR-REQ-001	Version	0.3
Status	Draft	Date	2026-10-03
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/vatreel		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-03	Amish Chadha	TRL 3 status on paper from VTR-CAL-001 for the constructable design; pit length added to the assumptions; R10 against the value-engineering target
0.2	2026-10-03	Amish Chadha	TRL 2 review; targets confirmed under Amish's pre-approval (VTR-DDR-001); R11 lift and R12 guarding added; concept status for each requirement
0.1	2026-09-30	Amish Chadha	Initial scaffold

VatReel requirements

Twelve measurable requirements. The status column gives the result on paper from the calculation note VTR-CAL-001 on the constructable design; every requirement is verified by test at TRL 4 or later. Ten are met on paper, R6 (corrosion) is at risk and R8 (portability) is not met as written.

Table 1. Requirements.

ID	Requirement	Target	Verification (TRL 4 or later)	Status on paper (VTR-CAL-001)
R1	Operator loads, runs and unloads a batch without any hand entering the liquor	0 hand contacts with liquor per batch	Observed trial with workers over at least 10 batches	Met on paper: the top carrier bar is 100 to 800 mm above the rim at every working depth; loading with long hooks
R2	Handle force to turn the loaded reel	100 N (22 lbf) or less with a 25 kg (55 lb) wet load	Spring balance at the handle on a water-filled test tank	Met: 90 N with the whole load on one bar
R3	Width of pits and vats served	1.0 to 2.5 m (3 to 8 ft), adjustable	Fit check on test frames of minimum and maximum width	Met: 1.0 to 2.52 m
R4	Immersion depth of the lowest carrier bar below the rim	Adjustable 0.3 to 1.0 m (1 to 3.3 ft)	Measurement at both settings	Met: any depth from 0.30 to 1.00 m
R5	Holding brake	Holds the rated load at any position with 1.5 times rated load applied, no slip	CalRig proof-load test	Met on paper: ratchet stresses under 13 MPa
R6	Corrosion resistance of wetted parts	No structural corrosion after 500 h immersion in sample liming, pickling and chrome liquors	Coupon immersion test and inspection	At risk: shown only by material choice; stainless chain 2 in chloride pickle is the weak point
R7	Heat tolerance for dye use	Operates in baths up to 98 °C (208 °F) with no deformation of reel or carriers	Hot-bath trial with temperature log	Met on paper: all materials rated
R8	Portability	Each module 25 kg (55 lb) or less; two people set up in 15 min or less	Weigh modules; timed setup trial	Not met: two modules 27.1 and 29.6 kg (two-person carries, 14.8 kg each at most); setup about 29 min
R9	Splash control at the crank position	No liquor droplets reaching the operator at normal cranking speed	Indicator paper on the operator's apron during trials	Met on paper: every sight line from the reel to the eye is blocked
R10	Cost	Value-engineering target USD 1,500 for prototype parts	Costed bill of materials	USD 1,437, USD 63 under the value-engineering target
R11	Lift and depth setting	Self-locking: holds the reel at any height with no brake applied; 100 N or less at the handwheel at 1.5 x load; lifts every carrier bar above the rim	Proof-load and handwheel force test	Met on paper: self-locking thread; 79 N; lowest bar 149 mm above the rim when lifted
R12	Guarding of moving parts	Chains, sprockets and ratchet enclosed; no nip point reachable from the operator position	Inspection against ISO 13857 by a competent person	Met on paper by enclosure; openings to check

■ Assumptions

- Pit rims are strong enough to carry the stands and load (under 600 N per stand); to be checked on site.
- Pits are at least 1.9 m (6.2 ft) long so the reel can swing; the arm pivot sits 350 mm in from the pit's end wall.
- The liquor stands about 150 mm below the rim; the rim is level with the floor.
- Hides can be toggled or hooked at one edge and yarn hanks looped over a bar so they stay on the reel as it turns.
- Existing process times and chemistry stay as they are; VatReel changes only how the load moves.
- A reel speed that a person can crank (about 5 rpm) is slow enough for even treatment.

Safety: These requirements cover moving machinery beside open pits of corrosive and hot liquor. Meeting them on paper does not make the design safe to use; gloves, eye protection and boots stay in use, and every site needs its own risk assessment.